AI Usage Declaration Form

Referencing the Output of Artificial Intelligence Language Models

Step 1: Select the declaration statement that best describes the level of AI tool usage in this assignment.

Table 1. Declaration Statement

Tick (✓) only 1 statement that applies	Declaration statements
	Declaration 1: We do not use any content generated by AI tools to be presented as our work.
	Declaration 2: We acknowledge the use of AI tool(s) to generate materials that aid with the process of completing this assessment (e.g. for background research or self-study). None of these materials are presented as our work.
✓	**Declaration 3:** We acknowledge the use of AI tool(s) to generate materials that are included in our work in modified forms.

Step 2: Complete the following form if you select either Declaration 2 or Declaration 3. You do not need to complete this section if you select Declaration 1.

Table 2. Details of AI Usage

Reference to one or more AI tools. Example: https://shorturl.at/8UT25	Claude Sonnet (Anthropic), accessed via https://claude.ai, April 2026.
List all prompts you have used.	Prompts used in this assessment covered the following categories: 1. "Help me understand what HD-level additions I need for my regression notebook for PRT564 Assessment 2" 2. "Review my Python notebook and identify gaps in statistical testing for HD criteria" 3. "Explain why Ridge regression is appropriate when Log_Population and Time_Index are correlated" 4. "Help me interpret the negative Time_Index coefficient and suggest a structural fix" All prompts were related to understanding, structuring, and improving our completed work. The core dataset, research questions, and analytical plan were developed independently in Assessment 1.

Reflect on how you have used these AI tools, how they have helped with your assignment and the limitations you encountered.	**How AI helped:** Claude (Anthropic) was used as an analytical assistant. It helped us identify gaps in our statistical testing (e.g. the need for Shapiro-Wilk normality testing and Wilcoxon signed-rank tests), helped us with reproducible scripts, and interpret results including the spurious negative Time_Index coefficient. **How we verified and modified output:** All code was run in our own Jupyter environment and every numerical result was verified against our actual dataset. Where AI-suggested code produced incorrect outputs (e.g. a misleading COVID percentage interpretation derived from Ridge standardised coefficients), we identified and corrected the error ourselves. All statistical interpretations were cross-checked against course materials. **Limitations encountered:** AI occasionally produced overconfident interpretations (e.g. claiming Ridge was definitively better than OLS when the M2→M3 improvement was not statistically significant at p=0.12). We corrected these and presented findings honestly. AI also initially modelled only two COVID structural periods; we identified the need for a third (Post_COVID_Recovery flag) through our own analysis of the per-capita trends in the data.
Do you maintain an accessible AI prompt history that is verifiable by your lecturer if requested?	**Yes.** Our full conversation history with Claude is accessible at https://claude.ai and can be provided to the lecturer upon request. Screenshots of key interactions are available if required.

Step 3: Answer the following question only if you selected Declaration 3. You do not need to complete this section if you select Declaration 1 or Declaration 2.

How have you modified the generative AI's output? You may include as many details as possible, including screenshots.

1. Code corrections: AI-generated Ridge regression code contained a misleading COVID impact percentage (16%) derived from standardised coefficients — mathematically incorrect when applied to a log-scale model. We identified this error, removed the incorrect interpretation, and replaced it with the raw data finding (47% demand reduction), which is statistically accurate and based directly on our dataset.

2. Structural decisions: AI initially modelled only two COVID periods (COVID / non-COVID). Through our own analysis of the per-capita trip trends in the data, we identified a third structural period (Post_COVID_Recovery, January 2022 onwards) where demand settled at a permanently lower baseline. We directed AI to implement this three-period structure; the conceptual insight was ours.

3. Model selection: The decision to use Ridge regression was grounded in our own observation of multicollinearity in the data (Log_Population and Time_Index both trend upward together). We used AI to understand the theoretical justification for Ridge in this context, but the modelling decision and its statistical rationale are based on our data analysis.

4. Python scripts: All Python scripts in the submission were reviewed line-by-line by team members. Code comments were verified against our understanding of the logic.

Student Signatures

Student ID	Full Name	Signature
S398425	**Nasla Maharjan**	
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S393829	**Suyog Kadariya**	

Date: 21st April, 2026

The contents of this AI Usage Declaration form have been adapted from:

- University of Birmingham (2025), *Acknowledging and Citing the Use of Generative AI by Student*, accessed on 4 Mar 2025. Link: https://www.birmingham.ac.uk/libraries/education-excellence/gai/acknowledging-gai-by-students
- University of Cambridge (2025), *Template Declaration of the Use of Generative Artificial Intelligence*, accessed on 4 Mar 2025. Link: https://www.cshss.cam.ac.uk/education/generative-artificial-intelligence-ai-and-scholarship/template-declaration-use-generative